Kok Yew NG (Mark NG), PhD

Reader and Head of MAvRiC, School of Engineering, Ulster University, UK

Adjunct Senior Research Fellow, School of Engineering, Monash University Malaysia

Vice-Chair, IEEE Control Systems (UK and Ireland Chapter)

EPSRC ILN+ Researcher in Residence, Digital Catapult

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Brief Biography

I received the BEng (Hons) in Electrical and Computer Systems Engineering and the Ph.D. in Fault Diagnosis and Control Systems from Monash University in 2006 and 2009, respectively. I am currently a Reader in Mechatronics Engineering and Control at the School of Engineering, Ulster University, UK, and I am attached to the Engineering Research Institute. I also currently lead the Multi-Agent and Advanced Robotics Centre (MAvRiC) at Ulster.

My research interests include fault diagnosis, mathematical modelling, digital twin, and data analytics for anomaly detection and classification.

In 2014–2015, I was a postdoctoral researcher at the Division of Vehicular Systems, Linköping University, Sweden, where I worked with Volvo Car Corporation (VCC) on advanced fault diagnosis schemes in vehicular engines using model-based and data-driven methods. For this research, I was instrumental in developing a Digital Twin/Simulation Testbed for realistic simulation and testing of residuals generation and fault diagnosis methods.

This research work was published in the *IEEE Control Systems Magazine* (click here) and the Digital Twin/Simulation Testbed can be downloaded via the main hosting site (click here) or its mirror at Linköping University (click here).

Throughout my career, I have secured more than £10.2 million in research grants and £23.1 million in infrastructure funding from various funders such as the Engineering and Physical Sciences Research Council (EPSRC), UK Research and Innovation (UKRI), Global Challenges Research Fund (GCRF), Aerospace Technology Institute (ATI), the British Academy, and the Northern Ireland Department for the Economy (DfE) in the UK; the Fundamental Research Grant Scheme (FRGS), Exploratory Research Grant Scheme (ERGS), and EScienceFund from the Ministry of Higher Education in Malaysia; and industries such as Volvo Car Corporation in Gothenburg, Sweden.

Overall, I have successfully supervised no less than 2 postdoctorals, 8 PhDs, and 3 Master's by Research candidates.

I am also currently attached to the Digital Catapult as an awardee of the EPSRC Innovation Launchpad Network+ (ILN+) Researcher in Residence Scheme. This research project aims to develop an energy mapping Digital Twin technology that contributes towards net zero in wind turbine energy. This technology encompasses the entire energy lifecycle, from mining through storage to utilisation in Northern Ireland (NI). This project also involves collaboration with the Offshore Renewable Energy Catapult.

Other highlights include being a co-investigator in SAFEWATER (click here), a £5 million project funded by UKRI-GCRF, where I led the development and the optimisation of embedded algorithms to control low-cost water disinfection technologies used in the rural areas in South America.

In addition, during the COVID-19 pandemic, I led the Modelling and Forecast Task Force at Ulster (click here) where we worked with the Southern Health and Social Care Trust to provide analysis to the Government Specialist Modelling Response Expert Group (SMREG) in NI. The main purpose of the project was to validate and inform the SMREG as well as help governing bodies in NI to better plan for intervention measures and ultimately flatten the curve. I was also a member of the COVID-19 Task Force set up by the IEEE Region 8 community. In addition, I led a team of researchers and data scientists from Ulster and Queen's University Belfast (QUB) to work with the Incident Controller for the State Health Incident Control Centre and Deputy Chief Health Officer of the Department of Health in Western Australia to model the outbreak of COVID-19 on commercial cargo vessels that ported in Australia.

In 2026, I successfully organised and hosted the inaugural "Robots at Work and Play" event at the Northern Ireland Science Festival, which attracted more than 200 attendees. This academic-led and students-run event showcased the latest advancements in robotics and automation, highlighting their applications in various industries and everyday life. Other key event partners include Elite Electronic Systems Ltd (KTP partner), Cobots.ie, IEEE Control Systems Society (CSS) UK and Ireland Chapter (UK&I), as well as EPSRC Innovation Launchpad Network+.

I am a Senior Member of the IEEE and I am currently the Vice-Chair of the IEEE CSS UK&I. I am a Moderator for *IEEE TechRxiv*, Associate Editor for *IEEE Access*, Editor for *PeerJ Computer Science*, and Section Editor for *Sage Science Progress*. I am also an Adjunct Senior Research Fellow with Monash University Malaysia where I served as a Lecturer from 2009, and subsequently as Senior Lecturer till 2017.

In addition, I am member of a few key technological committees in NI. Namely, I sit on the Academic Panel of the UK Digital Twin Centre (UKDTC), based at the Digital Catapult in Belfast. Funded by the Belfast Region City Deal and Innovate UK, the £37.6 million UKDTC aims to make digital twins more accessible and meaningful in the UK. I am also Co-I and an Advisory Board Member on the Artificial Intelligence Collaboration Centre (AICC), which is a £16.3 million initiative led by Ulster in partnership with QUB and supported by Invest NI and the Department for the Economy NI. I am also a Technology Strategy Groups Member (Digital Factory) for the Advanced Manufacturing Innovation Centre (AMIC), which is a £100 million investment under the Belfast Region City Deal. Other than that, I am a Steering Member for the Northern Ireland High Performance Computing (NI-HPC) consortium, a £2 million UK Tier-2 National HPC facility funded by the EPSRC and jointly managed by QUB and Ulster.

Education

2006 Monash University, BEng (Hons) Electrical and Computer Systems Engineering

2009 Monash University, PhD (Control Engineering and Fault Diagnosis)
Thesis: *Advancements in Robust Fault Reconstruction Using Sliding Mode Observers*

Experience

Ulster University

- 2025–present Reader, Mechatronics Engineering and Control Systems
  Head of the Multi-Agent and Advanced Robotics Centre (MAvRiC)
- 2021–2025 Senior Lecturer, Mechatronics Engineering and Control Systems
- 2017–2021 Lecturer, Mechatronics Engineering and Control Systems

Linköping University, Sweden

- 2016 Visiting Researcher (with Volvo Cars) (3 months)
- 2014–2015 Postdoctoral Fellow, Division of Vehicular Systems (with Volvo Cars)

Monash University, Malaysia

- 2017–present Adjunct Senior Research Fellow
- 2016–2017 Senior Lecturer, Electrical and Computer Systems Engineering
- 2009–2016 Lecturer, Electrical and Computer Systems Engineering
- 2006–2009 Graduate Researcher and Teaching Assistant

Honours and Awards

2026 'Outstanding' awarded for KTP project with Elite Electronic Systems Ltd

2025 Letter of Commendation for Excellent Overall Module Satisfaction Rates from the Pro-Vice Chancellor (Academic Quality and Student Experience), Faculty of Computing, Engineering and The Built Environment, Ulster University

2025 Nomination for Learning and Teaching Award, Ulster University Students' Union

2024 Nomination for Learning and Teaching Award, Ulster University Students' Union

2020 Learning and Teaching Award, Ulster University Students' Union

2018 Erasmus+ Staff Mobility Program

2012 Monash University Malaysia PVC's Award for Excellence in Research, Round 1

2012 Letter of Commendation for Excellent Unit Evaluation Result from the Associate-Dean (Education), Faculty of Engineering, Monash University Australia

2011 Monash University Malaysia PVC's Award for Excellence in Teaching, Round 2

2011 Monash University Malaysia PVC's Award for Excellence in Teaching, Round 1

2010 Top 50 Best Units offered by Faculty of Engineering, Monash University Across All Campuses (ranked #22)

2010 Monash University Malaysia PVC's Award for Excellence in Teaching, Round 2

2010 Monash University Malaysia PVC's Award for Excellence in Teaching, Round 1

2009 Monash University Malaysia PVC's Award for Excellence in Teaching, Round 2

2007 Degree by Research Scholarship for Ph.D. in Engineering

2006 Postgraduate Research Scholarship for Master of Engineering Science by Research

2002 Monash University Malaysia Entrance Scholarship

Professional Memberships

IEEE

- 2024–present Vice-Chair, IEEE UK and Ireland Control Systems Chapter
- 2022–2024 Secretary, IEEE UK and Ireland Control Systems Chapter

2020–present	Senior Member, IEEE
2010–2011	Auditor, IEEE Robotics and Automation Society (RAS) Malaysia Chapter
2009–2019	Member, IEEE
2018–present	Fellow, Higher Education Academy UK
2005–present	Graduate Member, Board of Engineers Malaysia (BEM)

Committees

2025–present	Academic Panel, UK Digital Twin Centre (UKDTC), Digital Catapult, NI
2025–present	Advisory Board Member, Artificial Intelligence Collaboration Centre (AICC), NI
2024–present	Technology Strategy Groups Member (Digital Factory), Advanced Manufacturing Innovation Centre (AMIC), NI
2024–present	Steering Member, Northern Ireland High Performance Computing (NI-HPC)
2023–2025	Athena Swan Champion, School of Engineering
2011	Faculty Representative, Campus Review Panel for Higher Degree by Research Course

Research Leadership and Activities

2024–present	Multi-Agent and Advanced Robotics Centre (MAvRiC) for Control, Digital Twin, and Adv. Manufacturing *Lab Head and Ulster Lead at School of Engineering, Ulster University*
2021–present	Control Systems, Fault Diagnosis, Data Analytics Using ML and DL for IIoT and Industry 4.0 Applications *Ulster Lead, collab. with Technical University of Applied Sciences Augsburg and Technology Transfer Center for Flexible Automation in Nördlingen, Germany*
2020–2022	Modelling and Predicting the Health Impact and Duration of SARS-CoV-2 Outbreaks on Cargo Vessels *Ulster Lead, collab. with Deputy Chief Health Officer of the Western Australia Department of Health*
2020–2022	COVID-19 Modelling and Forecast Task Force *Ulster Lead, collab. with NI Government Specialist Modelling Response Expert Group (SMREG)*
2019–2021	Identification and Classification of Multiple Weed Rice Species Using Mobile Computing *Ulster Lead, collab. with Monash University, Malaysia*
2015–2018	Mobile Control of Intelligent Lighting Systems *Monash Lead, collab. with ItraMAS Corporation Malaysia*

Publications: Preprint Articles

[1] S. Wucherer, R. McMurray, **K. Y. Ng**, and F. Kerber, *Learning to Predict Grip Quality from Simulation: Establishing a Digital Twin to Generate Simulated Data for a Grip Stability Metric*, 2023. arXiv: 2302.03504. [Online]. Available: https://arxiv.org/abs/2302.03504

Publications: Peer-Reviewed Journal Articles

[1] M. Hincapie et al., "Automated household-based water disinfection system for rural communities: Field trials and community appropriation," *Water Research*, p. 123 888, 2025. DOI: 10.1016/j.watres.2025.123888

[2] T. Fairooz, S. E. McNamee, D. Finlay, **K. Y. Ng**, and J. McLaughlin, "Enhancing Sensitivity of Point-of-Care Thyroid Diagnosis via Computational Analysis of Lateral Flow Assay Images Using Novel Textural Features and Hybrid-AI Models," *Biosensors*, vol. 14, no. 12, 2024. DOI: 10.3390/bios14120611

[3] O. Escalona, N. Cullen, I. Weli, N. McCallan, **K. Y. Ng**, and D. Finlay, "Robust arm impedocardiography signal quality enhancement using recursive signal averaging and multi-stage wavelet denoising methods for long-term cardiac contractility monitoring armbands," *Sensors*, vol. 23, no. 13, p. 5892, 2023. DOI: 10.3390/s23135892

[4] T. Fairooz, S. E. McNamee, D. Finlay, **K. Y. Ng**, and J. McLaughlin, "A novel patches-selection method for the classification of point-of-care biosensing lateral flow assays with cardiac biomarkers," *Biosensors and Bioelectronics*, vol. 223, p. 115 016, 2023. DOI: 10.1016/j.bios.2022.115016

[5] N. McCallan, S. Davidson, **K. Y. Ng**, P. Biglarbeigi, D. Finlay, B. L. Lan, and J. McLaughlin, "Epileptic multi-seizure type classification using electroencephalogram signals from the Temple University Hospital Seizure Corpus: A review," *Expert Systems with Applications*, p. 121 040, 2023. DOI: 10.1016/j.eswa.2023.121040

[6] **K.Y. Ng**, T. A. Codreanu, M. M. Gui, P. Biglarbeigi, D. Finlay, and J. McLaughlin, "Development of a mathematical model to predict the health impact and duration of SARS-CoV-2 outbreaks on board cargo vessels," *WMU Journal of Maritime Affairs*, 2022. DOI: 10.1007/s13437-022-00291-1

[7] P. Biglarbeigi, **K. Y. Ng**, D. Finlay, R. Bond, M. Jing, and J. McLaughlin, "Sensitivity analysis of the infection transmissibility in the UK during the COVID-19 pandemic," *PeerJ*, vol. 9, e10992, 2021. DOI: 10.7717/peerj.10992

[8] T. D. Do, M. M. Gui, and **K. Y. Ng**, "Assessing the effects of time-dependent restrictions and control actions to flatten the curve of COVID-19 in Kazakhstan," *PeerJ*, vol. 9, e10806, 2021. DOI: 10.7717/peerj.10806

[9] M. Jing et al., "COVID-19 Modelling by Time-varying Transmission Rate Associated with Mobility Trend of Driving via Apple Maps," *Journal of Biomedical Informatics*, p. 103 905, 2021. DOI: 10.1016/j.jbi.2021.103905

[10] L. J. Robertson et al., "Evaluation of the IgG antibody response to SARS CoV-2 infection and performance of a lateral flow immunoassay: cross-sectional and longitudinal analysis over 11 months," *BMJ Open*, vol. 11, no. 6, e048142, 2021. DOI: 10.1136/bmjopen-2020-048142

[11] **K. Y. Ng**, E. Frisk, M. Krysander, and L. Eriksson, "A Realistic Simulation Testbed of a Turbocharged Spark-Ignited Engine System: A Platform for the Evaluation of Fault Diagnosis Algorithms and Strategies," *IEEE Control Systems Magazine*, vol. 40, pp. 56–83, 2 2020. DOI: 10.1109/MCS.2019.2961793

[12] **K. Y. Ng** and M. M. Gui, "COVID-19: Development of a robust mathematical model and simulation package with consideration for ageing population and time delay for control action and resusceptibility," *Physica D: Nonlinear Phenomena*, vol. 411, p. 132 599, 2020. DOI: 10.1016/j.physd.2020.132599

[13] D. Jung, **K. Y. Ng**, E. Frisk, and M. Krysander, "Combining model-based diagnosis and data-driven anomaly classifiers for fault isolation," *Control Engineering Practice*, vol. 80, pp. 146–156, 2018. DOI: 10.1016/j.conengprac.2018.08.013

[14] L. H. Lee et al., "Sustainable approach to biotransform industrial sludge into organic fertilizer via vermicomposting: A mini-review," *Journal of Chemical Technology & Biotechnology*, vol. 93, no. 4, pp. 925–935, 2018. DOI: 10.1002/jctb.5490

[15] S. J. W. Tang, V. Kalavally, **K. Y. Ng**, C. P. Tan, and J. Parkkinen, "Real-Time Closed-Loop Color Control of a Multi-Channel Luminaire Using Sensors Onboard a Mobile Device," *IEEE Access*, vol. 6, pp. 54 751–54 759, 2018. DOI: 10.1109/ACCESS.2018.2872320

[16] J. H. T. Ooi, C. P. Tan, S. Nurzaman, and **K. Y. Ng**, "A Sliding Mode Observer for Infinitely Unobservable Descriptor Systems," *IEEE Transactions on Automatic Control*, vol. 62, no. 7, pp. 3580–3587, 2017. DOI: 10.1109/TAC.2017.2665699

[17] S. Tang, V. Kalavally, **K. Y. Ng**, and J. Parkkinen, "Development of a prototype smart home intelligent lighting control architecture using sensors onboard a mobile computing system," *Energy and Buildings*, vol. 138, pp. 368–376, 2017. DOI: 10.1016/j.enbuild.2016.12.069

[18] J. Y. Ng, C. P. Tan, H. Trinh, and **K. Y. Ng**, "A common functional observer scheme for three systems with unknown inputs," *Journal of the Franklin Institute*, vol. 353, no. 10, pp. 2237–2257, 2016. DOI: 10.1016/j.jfranklin.2016.03.020

[19] J. Y. Ng, C. P. Tan, **K. Y. Ng**, and H. Trinh, "New results in common functional state estimation for two linear systems with unknown inputs," *International Journal of Control, Automation and Systems*, vol. 13, no. 6, pp. 1538–1543, 2015. DOI: 10.1007/s12555-014-0315-x

[20] J. H. T. Ooi, C. P. Tan, and **K. Y. Ng**, "State and Fault Estimation For Infinitely Unobservable Descriptor Systems Using Sliding Mode Observers," *Asian Journal of Control*, vol. 17, no. 4, pp. 1458–1461, 2015. DOI: 10.1002/asjc.1033

[21] C. Y. Kee, C. P. Tan, **K. Y. Ng**, and H. Trinh, "New results in robust functional state estimation using two sliding mode observers in cascade," *International Journal of Robust and Nonlinear Control*, vol. 24, no. 15, pp. 2079–2097, 2014. DOI: 10.1002/rnc.2973

[22] **K. Y. Ng**, C. P. Tan, and D. Oetomo, "Disturbance decoupled fault reconstruction using cascaded sliding mode observers," *Automatica*, vol. 48, no. 5, pp. 794–799, 2012. DOI: 10.1016/j.automatica.2012.02.005

[23] **K. Y. Ng**, C. P. Tan, R. Akmeliawati, and C. Edwards, "Disturbance decoupled fault reconstruction using sliding mode observers," *Asian Journal of Control*, vol. 12, no. 5, pp. 656–660, 2010. DOI: 10.1002/asjc.231

[24] **K. Y. Ng**, C. P. Tan, Z. Man, and R. Akmeliawati, "New results in disturbance decoupled fault reconstruction in linear uncertain systems using two sliding mode observers in cascade," *International Journal of Control, Automation and Systems*, vol. 8, no. 3, pp. 506–518, 2010. DOI: 10.1007/s12555-010-0303-8

[25] **K. Y. Ng**, C. P. Tan, C. Edwards, and Y. C. Kuang, "New results in robust actuator fault reconstruction for linear uncertain systems using sliding mode observers," *International Journal of Robust and Nonlinear Control*, vol. 17, no. 14, pp. 1294–1319, 2007. DOI: 10.1002/rnc.1170

Publications: Peer-Reviewed Conference Articles

[1] **K. Y. Ng**, "Digital Twin of a Turbocharged Spark-Ignited Engine System for Control and Fault Diagnosis," in *The International Transport Forum (ITF) 2025 Summit on Transport Resilience to Global Shocks*, 2025.

[2] S. Kogileru, M. McBride, Y. Bi, and **K. Y. Ng**, "A Robust Tolerance Optimisation Framework for Automated Optical Inspection (AOI) in Semiconductor Manufacturing," in *DTNet+ Special Interest Group (SIG) Conference*, 2025.

[3] S. Kogileru, M. McBride, Y. Bi, and **K. Y. Ng**, "Design and Development of a Robust Tolerance Optimisation Framework for Automated Optical Inspection in Semiconductor Manufacturing," in *2025 IEEE 23rd International Conference on Industrial Informatics (INDIN)*, 2025, pp. 1–4. DOI: 10.1109/INDIN64977.2025.11279663

[4] S. Wucherer, R. McMurray, **K. Y. Ng**, and F. Kerber, "Predicting Maximum Permitted Process Forces for Object Grasping and Manipulation Using a Deep Learning Regression Model," in *8th IEEE Conference on Control Technology and Applications (CCTA)*, 2024, pp. 669–674. DOI: 10.1109/CCTA60707.2024.10666569

[5] N. McCallan, S. Davidson, **K. Y. Ng**, P. Biglarbeigi, D. Finlay, B. L. Lan, and J. McLaughlin, "Rebalancing Techniques for Asynchronously Distributed EEG Data to Improve Automatic Seizure Type Classification," in *2023 57th Annual Conference on Information Sciences and Systems (CISS)*, 2023, pp. 1–6. DOI: 10.1109/CISS56502.2023.10089669

[6] **K. Y. Ng**, "Digital Twins and Simulation Testbeds in Biomedical Engineering: From Mathematical Modelling and Classification for Diagnostics to Digital Simulation of Manufacturing Processes," in *Northern Ireland Biomedical Engineering Society Annual Symposium (NIBES) 2022*, 2022.

[7] S. Davidson, N. McCallan, **K. Y. Ng**, P. Biglarbeigi, D. Finlay, B. L. Lan, and J. McLaughlin, "Epileptic Seizure Classification Using Combined Labels and a Genetic Algorithm," in *2022 IEEE 21st Mediterranean Electrotechnical Conference (MELECON)*, 2022, pp. 430–435. DOI: 10.1109/MELECON53508.2022.9843099

[8] S. Davidson, N. McCallan, **K. Y. Ng**, P. Biglarbeigi, D. Finlay, B. L. Lan, and J. McLaughlin, "Seizure Classification Using BERT NLP and a Comparison of Source Isolation Techniques with Two Different Time-Frequency Analysis," in *2022 IEEE Signal Processing in Medicine and Biology Symposium (SPMB)*, 2022, pp. 1–7. DOI: 10.1109/SPMB55497.2022.10014769

[9] N. McCallan, S. Davidson, **K. Y. Ng**, P. Biglarbeigi, D. Finlay, B. L. Lan, and J. McLaughlin, "Seizure Classification of EEG based on Wavelet Signal Denoising Using a Novel Channel Selection Algorithm," in *2021 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, 2021, pp. 1269–1276.

[10] **K. Y. Ng**, E. Frisk, and M. Krysander, "Design and Selection of Additional Residuals to Enhance Fault Isolation of a Turbocharged Spark Ignited Engine System*," in *2020 7th International Conference on Control, Decision and Information Technologies (CoDIT)*, vol. 1, 2020, pp. 76–81. DOI: 10.1109/CoDIT49905.2020.9263792

[11] N. McCallan, D. Finlay, P. Biglarbeigi, G. Perpiñan, M. Jennings, **K. Y. Ng**, J. McLaughlin, and O. Escalona, "Wearable Technology: Signal Recovery of Electrocardiogram From Short Spaced Leads in the Far-Field Using Discrete Wavelet Transform Based Techniques," in *2019 Computing in Cardiology (CinC)*, 2019, pp. 1–4. DOI: 10.23919/CinC49843.2019.9005868

[12] P. Biglarbeigi, D. McLaughlin, K. Rjoob, Abdullah, N. McCallan, A. Jasinska-Piadlo, R. Bond, D. Finlay, **K. Y. Ng**, A. Kennedy, and J. McLaughlin, "Early Prediction of Sepsis Considering Early Warning Scoring Systems," in *2019 Computing in Cardiology (CinC)*, 2019, pp. 1–4. DOI: 10.23919/CinC49843.2019.9005630

[13] D. Jung, **K. Y. Ng**, E. Frisk, and M. Krysander, "A combined diagnosis system design using model-based and data-driven methods," in *2016 3rd Conference on Control and Fault-Tolerant Systems (SysTol)*, 2016, pp. 177–182. DOI: 10.1109/SYSTOL.2016.7739747

[14] W. J. Lee, **K. Y. Ng**, C. L. Tan, and C. P. Tan, "Real-time face detection and motorized tracking using ScicosLab and SMCube on SoC's," in *2016 14th International Conference on Control, Automation, Robotics and Vision (ICARCV)*, 2016, pp. 1–6. DOI: 10.1109/ICARCV.2016.7838614

[15] S. J. W. Tang, **K. Y. Ng**, V. Kalavally, and J. Parkkinen, "Closed-loop color control of an RGB luminaire using sensors onboard a mobile computing system," in *2016 IEEE Student Conference on Research and Development (SCOReD)*, 2016, pp. 1–5. DOI: 10.1109/SCORED.2016.7810062

[16] W. C. Chew, **K. Y. Ng**, and B. H. Khoo, "ReCon-AVe: Remote Controlled Automobile Vehicle for Data Mining and Analysis," in *2015 IEEE 39th Annual Computer Software and Applications Conference*, vol. 2, 2015, pp. 569–574. DOI: 10.1109/COMPSAC.2015.170

[17] S. J. W. Tang, **K. Y. Ng**, B. H. Khoo, and J. Parkkinen, "Real-Time Lane Detection and Rear-End Collision Warning System on a Mobile Computing Platform," in *2015 IEEE 39th Annual Computer Software and Applications Conference*, vol. 2, 2015, pp. 563–568. DOI: 10.1109/COMPSAC.2015.171

[18] **K. Y. Ng**, C. P. Tan, and D. Oetomo, "Enhanced fault reconstruction using cascaded sliding mode observers," in *2012 12th International Workshop on Variable Structure Systems*, 2012, pp. 208–213. DOI: 10.1109/VSS.2012.6163503

[19] C. Fernandes, **K. Y. Ng**, and B. H. Khoo, "Development of a convenient wireless control of an autonomous vehicle using apple iOS SDK," in *TENCON 2011 - 2011 IEEE Region 10 Conference*, 2011, pp. 1025–1029. DOI: 10.1109/TENCON.2011.6129266

[20] **K. Y. Ng** and C. P. Tan, “New results in disturbance decoupled fault reconstruction in linear uncertain systems using two sliding mode observers in cascade,” in *7th IFAC Symposium on Fault Detection, Supervision and Safety of Technical Processes*, vol. 42, 2009, pp. 780–785. DOI: 10.3182/20090630-4-ES-2003.00128

[21] **K. Y. Ng**, C. P. Tan, R. Akmeliawati, and C. Edwards, “Disturbance Decoupled Fault Reconstruction using Sliding Mode Observers,” in *17th IFAC World Congress*, vol. 41, 2008, pp. 7215–7220. DOI: 10.3182/20080706-5-KR-1001.01221

[22] **K. Y. Ng**, C. P. Tan, C. Edwards, and Y. C. Kuang, “New result in robust actuator fault reconstruction with application to an aircraft,” in *2007 IEEE International Conference on Control Applications*, 2007, pp. 801–806. DOI: 10.1109/CCA.2007.4389331

[23] **K. Y. Ng**, C. P. Tan, and R. Akmeliawati, “Tolerance towards sensor failures: an application to a double inverted pendulum,” in *Third IEEE International Workshop on Electronic Design, Test and Applications (DELTA'06)*, 2006, 6 pp.–434. DOI: 10.1109/DELTA.2006.92

Publications: Book Chapters

[1] M. M. M. Islam, J. I. Emon, **K. Y. Ng**, A. Asadpour, M. M. R. A. Aziz, M. L. Baptista, and J.–M. Kim, “Artificial intelligence in smart manufacturing: Emerging opportunities and prospects,” in *Artificial Intelligence for Smart Manufacturing and Industry X.0*. Springer Nature Switzerland, 2025, pp. 9–36. DOI: 10.1007/978-3-031-80154-9_2

Publications: Technical Report and Thesis

[1] **K. Y. Ng**, “Design and Development of a Simulation Environment and a Fault Isolation Scheme on a Volvo VEP4 MP Engine,” Research and Development Centre, Volvo Car Corporation, Gothenburg, Sweden, Tech. Rep., 2015.

[2] **K. Y. Ng**, “Advancements in robust fault reconstruction using sliding mode observers,” Ph.D. dissertation, Monash University, 2009. DOI: 10.4225/03/587c001b22509

Grants and Funding (Total $>$ GBP33mil)

2026 *Co-I*; GBP 199,000; Invest Northern Ireland Proof of Concept Grant (INI PoC)
Armband Continuous Cardiac Output and Rhythm Diagnostics (ACCORD) Device Technology and Commercialisation Consolidation

2026 *Co-I*; GBP 5,400,000; Aerospace Technology Institute (ATI)
Advanced Composites Design and Manufacturing Centre

2026 *Co-I*; GBP 4,600,000; Marie Skłodowska-Curie Actions (MSCA)
Advanced Monitoring For A Resilient Environment (AMORE) – Nano Sensing And Advanced Hybrid Modelling For Micro- And Nano-plastics, PFAs And Pathogens Detection

2026 *Co-I*; GBP 358,187; DfE Higher Education Research Capital Fund (HERC)
Decentralised Trusted Research Environment (D-TRE)

2026 *PI*; GBP 9900; British Academy Horizon Europe Pump Priming collaboration between UK and EU partners
SMART-COLLAB: Smart Collaborative Manufacturing through AI-Enhanced Human-Robot Cooperation and Digital Twin Technologies

2025 *PI*; GBP 4000; North West Tertiary Education Cluster (NWTEC)
Ulster University–Atlantic Technical University Mobility Initiative

2024 *Co-I*; GBP 16,300,000; Invest NI and Department for the Economy NI
Artificial Intelligence Collaboration Centre (AICC)

2024 *PI*; GBP 233,379; DfE Higher Education Research Capital Fund (HERC)
Multi-Agent and Advanced Robotics Centre (MAvRiC)

2024 *PI*; GBP 49,666; EPSRC Innovation Launchpad Network+ (ILN+) Researcher in Residence Scheme (RiR)
Empowering Green Futures: Developing Energy Mapping Digital Twin Technology for Sustainable Wind Turbine Energy in NI

2024 *Co-I*; GBP 29,000; InterTrade Ireland Innovation Boost
Industrial Partner: ARQ Reliability

2024 *PI*; GBP 214,060; Innovate UK: Knowledge Transfer Partnerships (KTP)
Industrial Partner: Elite Electronic Systems Ltd

2024 *Co-I*; GBP 21,450; Garfield Weston Trust (GWT)
Showcasing and Empowering Women in STEM

2023 *Co-I*; GBP 782,502; EPSRC
Multi-User Vector Network Analysis (VNA) Facility Core Equipment

2020	*Co-I*; MYR 980,000; Monash University Malaysia-ASEAN Sustainable Development Research Grant Scheme *Genomic variations and association study of agronomic traits in Malaysia, Myanmar and Thailand weedy rice (Oryza sativa* L.*)*
2019	*Co-I*; GBP 18,750; InterTradeIreland FUSION Industrial Partner: TERRA NutriTECH
2018	*Co-I*; GBP 4,889,812; EPSRC *SAFEWATER*
2018	*PI*; GBP 934.45; Erasmus+ Staff Mobility Programme Academic Partner: Tech. Uni. of Applied Sciences Augsburg, Germany
2018	*PI*; GBP 500; NVIDIA GPU Grant Programme
2015	*Co-I*; SEK 960,000; Volvo Cars, Gothenburg, Sweden *Design and Development of a Simulation Environment and a Fault Isolation Scheme on a VEP4 Engine*
2015	*Co-I*; MYR 127,000; Fundamental Research Grant Scheme, Ministry of Higher Education, Malaysia *Towards a Model Synthesis Framework for Conceptual Modeling of Cyber-Physical Systems*
2014	*Co-I*; MYR 93,200; Fundamental Research Grant Scheme, Ministry of Higher Education, Malaysia *A New Mathematical Framework for Smooth Curve Generation with Direct Incorporation of Uncertainties*
2014	*Co-I*; MYR 168,000; EScienceFund, Ministry of Higher Education, Malaysia *Design And Development of a Novel Ankle Rehabilitation Robot using Shape Memory Alloy Actuator*
2013	*PI*; MYR 50,000; ItraMAS Corporation Malaysia *Mobile Control of Intelligent Lighting Systems*
2012	*PI*; MYR 55,000; Monash University Malaysia Internal Grant *Application of Sliding Mode Observer for Fault Diagnosis on a Robotic System*
2012	*Co-I*; MYR 50,000; Exploratory Research Grant Scheme, Ministry of Higher Education, Malaysia *Working Model of a Parallel Pneumatic Regenerative Braking/Launch Assist for Light Motor Vehicles*
2010	*Co-I*; MYR 30,000; Fundamental Research Grant Scheme, Ministry of Higher Education, Malaysia *Development of Observer Schemes Using Sliding Mode Techniques for Robust State Estimation and FDI*
2010	*Co-I*; MYR 10,000; Fundamental Research Grant Scheme, Ministry of Higher Education, Malaysia *Effects of Noise Generation, Vibration and Model Size Reduction on TRSC in Solving Vehicle Noise Problems*
2010	*PI*; MYR 35,000; Monash University Malaysia Internal Grant *Embedded Mathematical, Simulation and Control Application on Mobile Computing Platform*

*GBP1 $\approx$ MYR5.46, SEK12.95 as of June 12, 2026

Journal Editorial Boards, Conference Organising Committee, and Society Activities

2018–present	*IEEE Access*, Associate Editor
2020–present	*IEEE TechRxiv*, Moderator
2020–present	*PeerJ Computer Science*, Editor
2025–present	*Sage Science Progress*, Section Editor
2025	*IEEE Power Electronics Society (PELS) UK and Ireland Chapter Best Thesis Award 2025*, Co-Chair
2025	*IEEE Signal Processing in Medicine and Biology Symposium (SPMB 2025)*, Tech. Prog. Chair & Lecture Chair
2025	*9th International Conference on Environment Friendly Energies and Applications (EFEA 2025)*, Int. Sci. Comm.
2024	*IEEE Signal Processing in Medicine and Biology Symposium (SPMB 2024)*, Tech. Prog. Chair
2024	*The 8th IEEE Conference on Control Technology and Applications (CCTA 2024)*, Workshop Chair & Session Chair
2024	*The 35th Irish Signals and Systems Conference (ISSC 2024)*, Prog. Comm. & Session Chair
2023	*IEEE Signal Processing in Medicine and Biology Symposium (SPMB 2023)*, Tech. Prog. Chair
2022–2023	*AIMS Mathematics*: SI on "Fault Diagnosis: Mathematical Models, Algorithms, & Application", Lead Guest Editor
2022	*IEEE Signal Processing in Medicine and Biology Symposium (SPMB 2022)*, Prog. & Lecture Chair
2020	*7th International Conference on Control, Decision and Information Technologies (CoDIT'20)*, Prog. Comm. Member
2020	*International Conference on Recent Innovations in Engineering and Technology (ICRIET-20)*, Prog. Comm. Member
2012	*International Conference on Intelligent Robotics, Automation and Manufacturing (IRAM 2012)*, Co-Chair of Reg.

Invited Lectures, Seminars, and Workshops

2026 "Digital Twins, Fault Diagnosis, and Simulation Testbeds: From Automotive Engines to Advanced Manufacturing"
Irish Control Systems Colloquium 2026 (IFACx event), ATU Sligo, Ireland

2025 "Digital Twins: More Than Just Hyped-Up Buzzwords"
School of Engineering, Monash University, Malaysia

2024 "Digital Twin of a Vehicular Engine for Fault Diagnosis"
Intelligent Systems Research Centre (ISRC), Ulster University, UK

2024 "Digital Twin of a Vehicular Engine as a Simulation Environment Platform for Fault Diagnosis"
Workshop, *The 8th IEEE Conference on Control Technology and Applications (CCTA)*, UK

2022 "Understanding Transmission Dynamics of Infectious Diseases Using Complex Networks"
School of Engineering, University of Warwick, UK

2020 "Engineering in Medical and Healthcare"
School of Mechanical, Aerospace and Automotive Engineering, Coventry University, UK

2019 "A Realistic Simulation Testbed of A Vehicular Engine System"
School of Engineering Research Seminar Series, Ulster University, UK

2019 Panel Discussion on "Robots and Automated Systems"
IET NI Robotics League, Ulster University, UK

2018 "A Turbocharged Petrol Engine System as a Simulation Benchmark Model for Fault Diagnosis"
Faculty of Electrical Engineering, Technical University of Applied Sciences Augsburg, Germany

2018 "Design and Development of A Fault Isolation Scheme on A Vehicular Engine System"
Faculty of Electrical Engineering, Technical University of Applied Sciences Augsburg, Germany

2017 "Beyond Calls and Games: Utilising The Full Potentials of Smartphones"
TEDx Sunway University: The Untold Ideas, Malaysia

2016 "Design and Development of a Simulation Environment for Fault Isolation on an Engine System"
Centre for Automotive Research, National University of Malaysia, Malaysia

2015 "Using a Smartphone Monoscopic Camera for Real-Time Lane Detection and Rear-End Collision Warning"
Machine Vision and Pattern Recognition Laboratory (MVPR), Lapeenranta University of Technology, Finland

2015 "Real-Time Lane Detection and Rear-End Collision Warning System on A Mobile Computing Platform"
Computer Science School of Computing, University of Eastern Finland, Finland

2014 "Robust Fault Diagnosis Using Sliding Mode Observers"
Division of Vehicular Systems, Linköping University, Sweden

2014 "Robust Fault Reconstruction Using SMOs and Real-Time Image Processing on A Mobile Device"
Department of Electrical, Electronic and Systems Engineering, National University of Malaysia, Malaysia

2012 "Disturbance Decoupled Fault Reconstruction Using Multiple Sliding Mode Observers"
Department of Telecommunications, Electrical, Robotics and Biomedical Engineering,
Swinburne University of Technology, Australia

2011 "Fault Reconstruction Using Sliding Mode Observer: Application to an Aircraft"
National Defence University of Malaysia, Malaysia

2010 "Robust Fault Reconstruction Scheme Using Sliding Mode Observers In Cascade"
School of Engineering, Deakin University, Australia

In the News and Key Outreach Activities

2026 "Robots at Work and Play"
Northern Ireland Science Festival

2022 "Ulster University in the Vanguard of Materials and Manufacturing"
The Irish Times

2020 "Modelling the Transmission Dynamics of COVID-19"
YouTube — Ulster University Research & Impact

2020 "Ancora Imparo with Dr Mark Ng"
SYNC, Monash University, Malaysia

2019 "Electric Vehicles in the UK"
BBC Newsline (interview with Donna Traynor)

2014 "More than Phone Calls and Games: Utilising the Full Potential of Smartphones"
The Star Newspaper, Malaysia

2012 "Life in Academia"
POSTGraduan, Malaysia

2011 "Dr Mark Ng: A PhD and Beyond!"

Momentum, Sunway University, Malaysia

2009 "First engineering PhD has a connection with Sunway"
Blaze, Sunway University, Malaysia

2009 "First Engineering PhD for Sunway"
Monash Magazine

2009 "Doctor's Dream Fulfilled"
The Star Newspaper, Malaysia

Reviewer for Funding

2019–present Newton Funds

Reviewer for International Peer-Reviewed Journals

Automatica (Elsevier)
IEEE Transactions on Industrial Electronics (TIE) (IEEE)
IEEE Transactions on Instrumentation and Measurement (TIM) (IEEE)
IEEE Journal of Biomedical and Health Informatics (JBHI) (IEEE)
IEEE Access (IEEE)
International Journal of Robust and Nonlinear Control (IJRNC) (Wiley)
Control Engineering Practice (CONENGPRAC) (Elsevier)
European Journal of Control (EJCON) (Elsevier)
Asian Journal of Control (AJC) (Wiley)
Computers and Electrical Engineering (COMPELECENG) (Elsevier)
Circuits, Systems and Signal Processing (CSSP) (Springer)
Building Simulation (Springer)
International Journal of Applied and Computational Mathematics (IACM) (Springer)
International Journal of Advanced Robotic Systems (IJARS) (SAGE)
International Journal of Control (IJC) (Taylor & Francis)
Australian Journal of Electrical and Electronics Engineering (AJEEE) (Taylor & Francis)

External PhD Examiner

Deakin University, Australia
Coventry University, UK

Supervision of Graduate Research and Associates

2026	Mr Valerio Leone (Research Assistant, EU Erasmus+ Traineeship Programme)
2026	Mr David Traxler (Research Assistant, EU Erasmus+ Traineeship Programme)
2026	Mr Mathieu Garcia (Research Assistant)
2025–present	Mr Ranga Kanteti (PhD, Ulster University)
2025	Mr Martin Kučera and Mr Tomáš Smrčka (Research Assistant, EU Erasmus+ Traineeship Programme)
2024–2026	Mr Phil Watson (InterTradeIreland Research Associate, Ulster University and ARQ Reliability)
2024–2026	Ms Shruthi Kogileru (KTP Associate, Ulster University and Elite Electronic Systems Ltd)
2024–present	Mr Brian Kirch (PhD, Ulster University)
2021–present	Ms Stefanie Wucherer (PhD, Ulster University, Technical University of Applied Sciences Augsburg, and Technology Transfer Center for Flexible Automation in Nördlingen)
2021–present	Mr Will Aston (PhD, Ulster University)
2020–2024	Dr Towfeeq Fairooz (PhD, Ulster University)
2019–2024	Dr Scot Davidson (PhD, Ulster University)
2019–2024	Dr Niamh McCallan (PhD, Ulster University)
2019–2020	Mr Colin Maher (FUSION Project Manager, Ulster University and TERRA NutriTech)
2017–2020	Mr Da Yi Lee (Master's by Research, Monash University)
2015–2018	Mr Leong Hwee Lee (Master's by Research, Monash University)
2015–2018	Mr Samuel Jia Wei Tang (Master's by Research, Monash University)
2012–2016	Dr Jiunn Yea Ng (PhD, Monash University)
2011–2015	Dr Jeremy Hor Teong Ooi (PhD, Monash University)

2010–2014	Dr Chew Yee Kee (PhD, Monash University)
2009–2012	Dr Jen Nee Lim (PhD, Monash University)

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